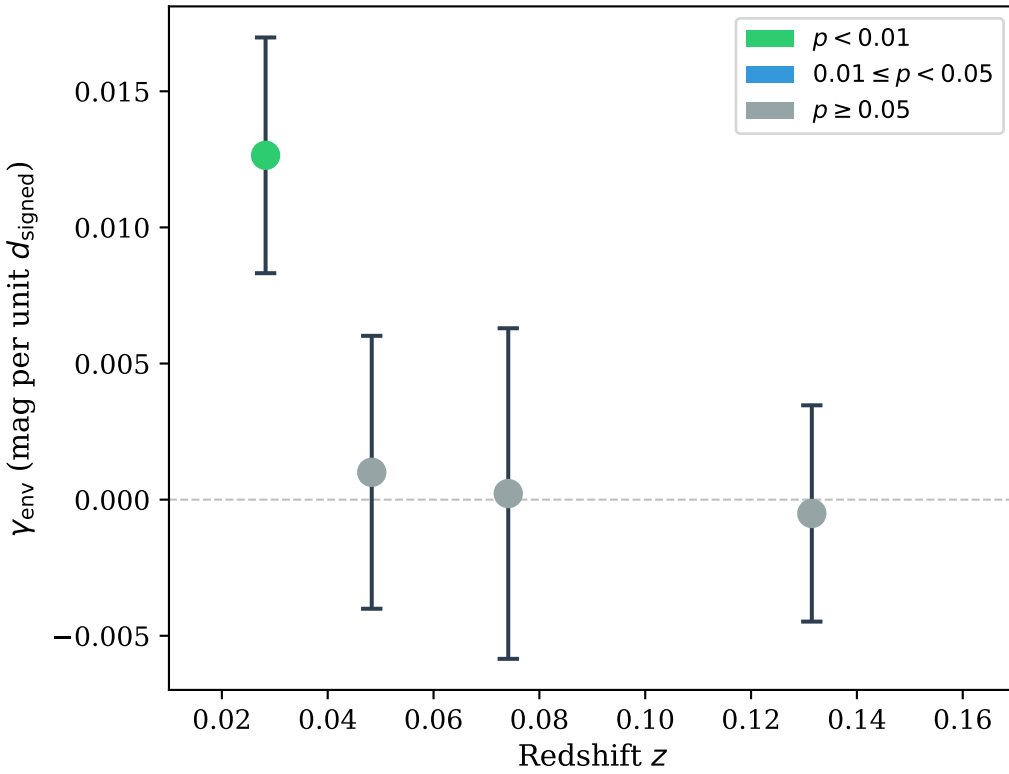
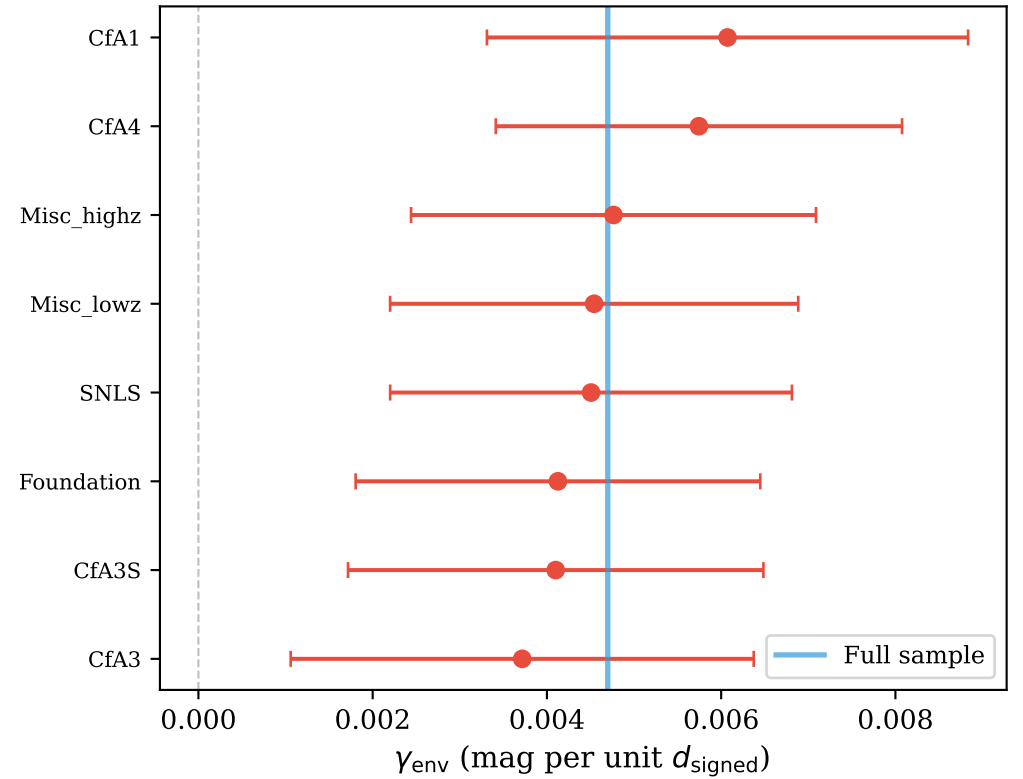


(a) Environment Signal vs. Redshift



(b) Leave-One-Survey-Out Stability



(c) Key Results Summary

KEY RESULTS: SN x VOID ENVIRONMENTCombined Sample ($z \leq 0.157$):REVOLVER: $\gamma = +0.0047 \pm 0.0023$, $p = 0.039$ VIDE: $\gamma = +0.0046 \pm 0.0026$, $p = 0.076$ VoidFinder: $\gamma = +0.0030 \pm 0.0018$, $p = 0.088$ NGC Footprint ($p < 0.01$ for two finders):REVOLVER NGC: $\gamma = +0.0103 \pm 0.0038$, $p = 0.006$ VIDE NGC: $\gamma = +0.0111 \pm 0.0040$, $p = 0.006$ Low- z Bin ($0.02 \leq z < 0.04$): $\gamma = +0.013 \pm 0.004$, $p = 0.003$ (most significant)

Robustness:

* 12/12 LOSO runs: all positive γ , all $p < 0.05$ * Survey FE: γ unchanged ($\Delta\gamma < 0.0005$)* Z-matched SGC: still null $\rightarrow$ footprint-specific

(d) Physical Interpretation

PHYSICAL INTERPRETATION

Observation:

SNe in/near cosmic voids appear ~ 0.01 mag
BRIGHTER than SNe in overdense regions,
at fixed LCDM distance.

MTDF Prediction (Section 2.3):

If the stress field has finite coherence
scale, large underdensities can sustain
different stress states $\rightarrow$ local G_{eff} .

Consistency Checks:

- [Y] Signal concentrated at low z
- [Y] Same sign across 3 void finders
- [Y] Independent of host-mass step
- [Y] Stable under survey fixed effects
- [Y] Not driven by any single survey

Caveat: Only 4 SNe strictly inside voids.

Inference is from continuous signed-distance.